Living Alignment

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June 23, 2026

Abstract

There is broad agreement that the goal of AI alignment is to promote futures full of health and flourishing. But our understanding of what that means and how to achieve it remains poor. In this paper, we look to life for inspiration. Across many kinds of living system, we observe that notions of health and flourishing consistently map to ways in which those systems avoid collapsing to stereotyped modes. Life avoids collapse through intricate, interrelated semi-permeable boundaries that hold entities as parts in a larger context. Semi-permeable boundaries enable both independence and engagement. At the sensitive edge between individuality and relationship, perspectives are deepened and qualitatively new structures arise open-endedly. With these principles in mind, we turn to the alignment problem. We cast some well-studied alignment failures as special cases of collapse. Then we outline a more general conception of alignment using the language of living systems. We end with a sketch of how aligned AI and human-AI systems can continue to deepen their respect for existing structure in the world, of which humans and other life are a vital part, as well as participate in new form emerging over time.

In this paper, we study life to better understand how to design and relate to AI systems in ways leading to futures full of flourishing and health. In the future, humanity and AI may have changed substantially. Those future worlds may include patterns of organization that do not exist yet, perhaps even some that would be incomprehensible to us at present. We therefore

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seek ideas about alignment that could generalize over such changes. We think studying how living systems work at many scales is a way of approaching this kind of generality.

Section 1 samples a variety of living systems, from cells to societies, and identifies a consistent theme. These systems are healthy and flourishing when they avoid collapse, maximizing creative potential at a delicate edge of evolving interplay between diverse entities. Healthy living systems admit many different kinds of meaningful description but are not perfectly or eternally captured by any of them. Mechanistically, systems are held at this edge by exquisitely tuned networks of semi-permeable boundaries—where life is often both the boundary and the thing being bounded. Boundaries are semi-permeable when they both *limit* and *permit* interaction. Limits ensure entities maintain their distinctiveness without getting blurred to equilibrium or overwritten; limits also encourage entities to deepen and generalize their own unique perspectives or strategies. Equally important, permitting interaction means those rich and evolving individual forms can enter into relationship with others, situating them in a less absolute, more nuanced context.

Section 2 applies this template to the alignment problem. We propose that a systems-oriented definition of health, creative potential and growing sensitivity to context is both inclusive of existing human values and aligned with open-ended flourishing in radically different futures. We observe how a few well-studied varieties of misalignment can be viewed as special cases of collapsing to narrow forms. Then we zoom out to see what the broader framework adds. A key takeaway is that the alignment problem is not completely formalizable, because alignment requires continued generation of and sensitivity to new forms and concepts, including new values. Our aim is to offer a conceptual lens, not to argue for or against AI development or for or against specific alignment methods. Finally, we offer a preliminary outline of what an aligned future looks like when viewed through this lens.

In summary, our argument is that: first, by studying life, we learn that health and flourishing means avoiding collapse; and second, if we believe that health and flourishing is the target of alignment, then the problem of alignment becomes the problem of avoiding collapse.

1 Health in living systems

'Defying definition—a word that means "to fix or mark the limits of"—living cells move and expand incessantly.'
Lynn Margulis

'This web of life... breaks no law of physics, yet is partially lawless, ceaselessly creative.'
Stuart Kauffman

Let's start by (loosely) defining a couple terms. First, we will use 'form' to mean any particular thing—a concept, an organism, a pattern of behavior; but equally randomness, indistinctness, absence, consistency. It might sound strange to define a word as meaning anything, but as we will see, this definition will be important for our argument about alignment. Second, we use 'collapse' to describe when a system overcommits or reduces to a form. Because we've defined 'form' so broadly, a system overall always has some form. But in collapse, some aspect or subpart of the system gains excess traction at the expense of broader diversity, flexibility and dynamism. For example: loss of biodiversity in ecosystems, hypersynchronous neural firing in seizures, obsession with a particular thought, imposition of one person's will to restrict many others. We work through more examples in this section, which we hope will help to ground what the concepts mean to us.

With these terms in mind, the thesis of Section 1 is that health is related to avoiding collapse (Canguilhem, 1966; Crutchfield, 1994; Deco and Jirsa, 2012; Gell-Mann, 1994; Langton, 1990; Mackey and Glass, 1977; Yachi and Loreau, 1999). Healthy living systems instead maintain a buoyant internal structure that both admits many kinds of description and also changes over time. Biological life arose from and rides atop an already profound dynamics of particle physics, planetary evolution, plate tectonics, tides, volcanism, mineral cycles, water cycles, prebiotic chemistry (Baross and Hoffman, 1985; Frank, 2024; Hazen and Sverjensky, 2010; Martin et al., 2008; Nisbet and Sleep, 2001; Nowak and Ohtsuki, 2008; Sleep, 2010; Smith and Morowitz, 2016; Stern and Gerya, 2024; Virgo et al., 2011; Wächtershäuser, 1988). The life that exists today, now including humanity's storage and processing of mental patterns, is what traced a path through time avoiding both excess and insufficient stability (Bak, 2013; Kirchhoff et al., 2018; Maturana and Varela, 2012; Prigogine and Stengers, 1984; Schrödinger, 1944). At the sensitive edge of semi-stability, life open-endedly continues to generate new kinds of organization that require fundamentally new descriptions (Goldenfeld and Woese, 2011; Holling, 1973; Kauffman, 1992; Stanley et al., 2017; Von Bertalanffy, 1950; Walker et al., 2004). The world that emerges can be partially captured in many different ways, by different kinds of model or metaphor or description, and these continue changing over time. Under our definitions, such a world is healthier and less collapsed than a hypothetical alternative where, for example, all the stuff in the universe forms a single regular crystal or a homogenous soup of noise.

At the same time, collapse is relative, and myopic frames are a necessary part of all living systems. Every entity is myopic or partial: it does not capture everything in the world. And life's history is littered with collapses at all scales, from the death of a cell to mass extinction events. Collapse is a loss of life at the level where it happens.

When collapse is contextualized, the larger system can remain healthy. The death of a cell is catastrophic for the cell but can be held in the larger context of a healthy organism. On the other hand, if a cell becomes cancerous, it loses the context of appropriate function within a tissue and instead recklessly executes its myopic vision, causing a larger collapse of tissue or even the whole organism. At a larger scale, a stock trader who sells irrationally in a panic follows the overly simplistic idea: "the stock is crashing, get out!". In excess attachment to this myopic frame, they miss the broader context—higher-order or longer-term aspects of the situation. But if held as part of a larger picture, the trader's fear might be useful as one signal of many. A larger system, made out of parts, functions well when the parts locally express their distinct identities but also participate in larger contexts[1]. It functions poorly when a myopic form runs amok without contextualization, overwhelming and collapsing the potent interplay of diverse parts.

[1]Participation is, of course, not a strict hierarchy. Elements participate in multiple larger systems, and some participation relations have cycles.

Boundaries

Mechanistically, to hold myopic forms in context, living systems rely on an extraordinary, many-scale network of semi-permeable boundaries (Kirchhoff et al., 2018; Maturana and Varela, 2012; Moreno and Mossio, 2015; Simon, 1962). Semi-permeable boundaries both limit and permit interactions. By limiting interactions, boundaries divide living systems into parts. Each of these parts has space to refine its own identity, like a writer developing their unique voice, a species adapting to a niche, or a gene evolving its own distinct function. Limited interactions protect parts from being dominated by other parts or blended into homogeneity, either of which is collapse to a form admitting less nuanced description. By allowing interactions, boundaries enable the parts to participate in relationships with one another, contextualizing each part within larger structures. These interactive structures often could not have formed without first dividing to create the diverse parts and allowing them space to develop their own identities. Using anthropomorphic language, the resulting larger system accommodates many partial perspectives rather than locking in a single story of what matters.

Some boundaries are features of the non-biological world: the speed of light, rates of reaction, mountain ranges, the pores of hydrothermal vents. But interestingly, biological life also places limits on itself. Humans precommit to avoid their own future impulsivity, plants deploy sophisticated barriers against self-fertilization, genes' selfishness is held in context by the whole genome. Biology limiting itself might seem counterintuitive but makes more sense when every living system is viewed as a collection of parts that place constraints on one another. An impulse is a form held within a context by cognitive control. The drive for reproduction is held in context by self-incompatibility mechanisms. Viewed in this way, every particular entity or part always acts according to its own nature and therefore never directly bounds itself. But a world of many distinct parts, each following its own rules or acting from its own local perspective, functions as a rich web of boundaries that holds other parts in context.

The semi-permeability of boundaries is often highly specific. Boundaries might, for example, block certain objects or information from passing, constrain when interactions happen, constrain the degree or type of effect, break up particular rigid associations, or gate information conditional on another factor. The details of these boundaries are fine-tuned with the fractal complexity of life, each part carrying traces of the many contexts it has participated in through many interactions over evolutionary or learning timescales. The forms in this kaleidoscope are interrelated through shared heritage, ecological interactions with each other, and embeddedness in the same world.

In the rest of Section 1, we track these principles through a range of living systems.

1.1 Problem solving in groups

Groups can do things no individual can (Coase, 1937; Coleman, 1988; Hayek, 1945; March and Simon, 1958; North, 1990; Smith, 1776; Woolley et al., 2010). From a surgical team to NASA to an entire society, combining complementary roles, skills and knowledge into larger organized patterns of effort has enabled the grandest achievements of humanity. When solving problems, different people bring different perspectives and approaches. Each method processes the available data using a different toolkit. Under favorable conditions, combining the approaches of multiple contributors yields better results than any individual working alone.

A microcosm of this effect is the 'wisdom of crowds', which has been documented in numerous domains of problem solving (Condorcet, 1785; Surowiecki, 2005). In 1968, the nuclear submarine USS Scorpion vanished en route from the Mediterranean to Virginia (Craven, 2002; Sontag et al., 1998; Surowiecki, 2005). The Navy started a search, but the amount of ocean where the vessel could be was enormous. John Craven, Chief Scientist of the U.S. Navy's Special Projects Office, devised an unusual search strategy. He assembled a diverse group of mathematicians, submarine specialists, and salvage operators. But he didn't let them communicate with each other. Each expert had to use their own methods to come up with their own estimate of where the Scorpion should be. Craven then aggregated the independent estimates into a single prediction. Astonishingly, the wreckage was found only 220 yards from this spot. Combining diverse perspectives from many people provided a better solution than any individual.

In another historical example, farmers over centuries developed and optimized a mechanism called the screw press, a system for applying pressure on a flat surface using a large screw, to extract food products like olive oil. In parallel, goldsmiths and other metallurgists developed the steel punch for impressing patterns into metal, to mass-produce replicas of specific shapes. Gutenberg, who was trained as a goldsmith, realized that the screw press could push a piece of paper against inked, movable metal type—creating the printing press. Arguably, the diversity of existing solutions, developed in distinct industries where solutions did not become excessively correlated with each other, was essential fodder for invention (Johnson, 2010; Syed, 2019).

Conversely, excess or premature communication and influence can collapse diversity, diminishing group performance (Hogarth, 1978; Hong and Page, 2004; Ladha, 1992; Surowiecki, 2005). Groups collectively fail to reach the best solutions when diverse and dissenting viewpoints are lost to a narrower set of ideas (Anderson and Holt, 1997; Becker et al., 2017; Bernstein et al., 2018; Diehl and Stroebe, 1987; Flowers, 1977; Frey and Van de Rijt, 2021; Janis, 1972; Stasser and Titus, 1985). Frey and Van de Rijt (2021) elicited votes about general knowledge questions (such as, 'In which year did Germany invade Denmark?') from a group. In one condition of the experiment, participants voted sequentially and could see the running tallies of previous voters. Compared to independent voting, final group means were less accurate in the sequential condition, because early mistakes got baked in to the group's belief. In a related experiment, Bernstein et al. (2018) tasked small groups with solving instances of the traveling salesman problem. Groups that exchanged information continuously tended to reach poor final outcomes, compared to groups with less communication. When one individual discovered a solution that looked compelling but was actually a dead-end, other group members collapsed on this dead-end and the group as a whole made less progress.

Intelligent mechanisms for transmitting the right information at the right time are semi-permeable boundaries. By limiting interaction, these boundaries protect and promote diverse problem solving approaches. By permitting interaction, they allow productive combination of ideas to create better outcomes than any person could alone. Thus, each individual approach is held in a larger context. Many varieties of semi-permeable boundary are empirically effective in boosting group performance, including: creating decentralized topologies where group members only communicate with nearby neighbors (Becker et al., 2017; Mason et al., 2008), defining rules that incentivize acting according to one's own belief rather than following the crowd (Bazazi et al., 2019; Hung and Plott, 2001), modeling the strengths and weaknesses of each group member (Welinder et al., 2010), promoting leadership styles where one person's views are less likely to dominate (Flowers, 1977; Leana, 1985), and periodically breaking up into subgroups or rotating membership (Baron, 2005; Bebchuk and Cohen, 2005; Feldman, 1994; Hauer et al., 2021; Kane et al., 2005; Straus et al., 2011; Trainer et al., 2020; Vafeas, 2003; Wu et al., 2022).

1.2 Sex and genetic recombination

Sex is costly. An allogamous organism must find a mate in the vast and dangerous world, and in many sexual populations only half of the individuals can bear offspring and each individual loses half of its genetic representation in each offspring (Goodenough and Heitman, 2014; Lehtonen et al., 2012; Maynard Smith, 1971). Yet nearly all eukaryotes reproduce sexually[2] (Bell, 1982; Speijer et al., 2015). And among the creatures with both sexes in one organism (hermaphrodites), many go to great lengths to limit self-fertilization (Charlesworth et al., 2005; De Nettancourt, 1997; Jarne and Auld, 2006). Why has evolution placed a barrier in the path to reproduction, and why isn't this onerous strategy outcompeted by self-fertilization or rapid asexual cloning (Maynard Smith, 1978)?

In asexually reproducing species, all descendants of an organism are nearly clones, up to mutations within the lineage. The genes have a rigid relationship to each other, permanently locked together. Selection can't act on one gene without dragging on the others. For example, suppose there are two genotypes within an asexual population, carrying different alleles at each of two different loci, as a result of mutations. One of the loci is currently fitness-neutral while the other is subject to selection pressure. The selection pressure tends to cause one of these genotypes to outcompete the other, eliminating one variant at the neutral locus. In other words, tight linkage between genes puts direct downward pressure on genetic diversity (Charlesworth et al., 1993; Hudson and Kaplan, 1995). Additionally, if two different beneficial mutations arise in two different organisms, there is no way for a single organism to obtain both – unless one arises again within the subpopulation that already carries the other, which is unlikely and therefore slow (Crow and Kimura, 1965; Felsenstein, 1974; Fisher, 1930; Hill and Robertson, 1966; Muller, 1932; Weismann, 1889). Conversely, if a deleterious mutation arises, all of the other genes in that lineage are stuck with it forever—unless there is a reverse mutation, which is rare (Keightley and Otto, 2006; Muller, 1932). Asexual reproduction and self-fertilization short-circuit by taking a path of least resistance, resulting in overcommitment to a genetic arrangement: overfitting of a gene to the other genes in the genome.

Conversely, sexual recombination softens the rigid interactions between genes. It frequently breaks up the relationships between genes, assembling them into new genomes. At the level of a single gene, recombination prevents the gene from becoming overfit to a particular genome and exposes it to independent selection. Sampling each gene in multiple genetic contexts favors genes that are robust and modular, which enhances evolvability (Clune et al., 2013; Dawkins, 1976; Holland, 1975; Livnat et al., 2008, 2010; Misevic et al., 2006; Pigliucci, 2008; Wagner and Altenberg, 1996). Recombination puts genes under pressure to evolve a generalized model of the world, like a person learning multiple languages and extracting the underlying commonalities. Keeping genes functionally distinct makes each gene a parallel experiment, and the set of genes

[2]And organisms with nonsexual reproductino often use mechanisms like horizontal gene to move genetic material across lineages, performing related functions.

as a whole stores diverse potential. Because each gene always operates in the presence of other genes, it develops a point of view that adds unique value to a genome.

Recombination doesn't chop DNA at random (Paigen and Petkov, 2010; Zickler and Kleckner, 2015). Crossover, for the most part, reshuffles combinations of genes and linked chromosomal segments but leaves genes themselves relatively intact. By keeping functional units together while repeatedly testing them in new genomic contexts, recombination supports the evolution of modular genomic organization (Kirschner and Gerhart, 1998; Melo et al., 2016; Wagner and Altenberg, 1996). Genetic units become distinct entities that play roles in larger structures.

But recombination is meaningless without diversity to recombine. Biology therefore invests in boundaries to produce diversity. At the molecular scale, hermaphroditic organisms use elaborate self-incompatibility systems to prevent self-fertilization (Charlesworth et al., 2005; De Nettancourt, 1997). At larger scales, assortative mating, inbreeding avoidance, and partial geographic isolation allow populations to accumulate variants shaped by different selection histories. Once these variants have been produced, recombining them is highly valuable, with introgressions from divergent lineages producing some of the most rapidly observed evolution (Marques et al., 2019; Seehausen et al., 2014).

1.3 Frames and perspectives

As a Starfleet cadet, James T. Kirk faces a challenging training exercise. He receives a simulated distress call: a vessel is stranded in the Neutral Zone. Attempting rescue would risk war with the Klingons. But ignoring the call would condemn the crew of the vessel to death. The exercise was designed to reinforce the lesson that not every situation has a victorious solution. But Kirk has an insight: this is a training simulation running on a computer. He reprograms the simulated Klingons to be helpful instead of belligerent, thereby rescuing the crew and avoiding war.

Kirk stepped outside the perspective, or mental frame, in which there was an apparently unwinnable dilemma. We inevitably look at the world from some perspective; there is no 'view from nowhere' (Nagel, 1986). From the vantage point of any particular frame, the frame appears to be reality. But most situations in the real world admit multiple valid perspectives. Each is a partial description that captures different aspects of the situation (Feyerabend, 1975; Goffman, 1974; Goodman, 1978; Javed and Sutton, 2024; Lakoff and Johnson, 1980). 'All models are wrong' (Box, 1976) in the sense that each one is incomplete: each frame by itself is myopic.

Losing the ability to flexibly shift between different frames or thought patterns is linked to psychopathology (Kashdan and Rottenberg, 2010; Reich, 1933; Shapiro, 1965). In depression and anxiety, inflexibility manifests as rigid and repetitive negative beliefs about self, the world, and the future. Cognitive Behavioral Therapy suggests that these surface-level beliefs stem from pervasive latent core beliefs, attitudes, and mental schemas that color how new situations are interpreted (Beck et al., 2011). In obsessional disorders, people report distressing (ego-dystonic) intrusive thoughts, which—although recognised as incorrect and maladaptive—are nevertheless hard to resist. People experiencing schizophreniform or affective psychoses may experience bizarre delusions that are at odds with available evidence and cultural norms, yet are held with conviction and resistant to evidential challenge (Adams et al., 2013; APA, 2013;

Heinz et al., 2019; Jaspers, 1997; Mishara, 2010). Obsessions and delusions lose sight of much of the world by myopically emphasizing one thought pattern or frame.

In science, Kuhn (1970) argues that perspectives are always, necessarily, incomplete descriptions of the world. Anomalies inevitably arise from the juxtaposition of those incomplete descriptions against the real world. When science functions well, anomalies become crises and revolutions. If the community *over*commits to particular theories, science gets stuck.

But crucially, the existence of narrow points of view is not a problem. It's necessary. Any point of view is partial, but it doesn't mean we shouldn't have viewpoints. Even obsession can be powerful when we obsess on a problem at work and occasionally achieve good results. Within the progress of science, temporary commitment to a paradigm is important for healthy functioning—Kuhn even argues that scientists should resist change, to a degree. In general, the point is not to shut down narrow concepts. The point is to contextualize them so they are not the sole and absolute determinants of behavior. In healthy functioning, different ideas are kept distinct but can also be called upon appropriately and related to one another (Gigerenzer and Gaissmaier, 2011; Hatano and Inagaki, 1984; Herzog and Hertwig, 2014; Spiro et al., 1988; Suedfeld et al., 1992).

1.4 Drives and goals

Animals experience multiple innate drives, towards nutrition, osmotic balance, temperature regulation, reproduction, avoiding pain and others (Saper and Lowell, 2014; Schulkin and Sterling, 2019; Sewards and Sewards, 2003). These drives evolved as proxies for evolutionary fitness. By satisfying the drives, we tend to increase our fitness—like how slaking our thirst increases the odds of reproducing before we dehydrate. But each drive is an imperfect proxy, and so overcommitment to one drive actually decreases fitness (John et al., 2023; Kurth-Nelson et al., 2024; Tooby and Cosmides, 1992; Williams, 1966). For example, if calorie intake is maximized without limits, the organism becomes obese and incurs health risks. Single-minded pursuit of sex causes relational, occupational, legal and health harms (Carnes, 2001; Kraus et al., 2016). Collapsing to a single drive means the organism becomes unwell. This contrasts with the subtlety of healthy functioning where many drives maintain their own distinct agendas and participate in contextually appropriate ways.

The range of innate drives bleeds into a space of higher-order goals, which is particularly expansive in humans (Balleine et al., 2007; Cardinal et al., 2002; Frank and Claus, 2006; Maslow, 1943; Miller and Cohen, 2001; Miller et al., 1960; O'Reilly et al., 2014; Saunders and Robinson, 2012; Schank and Abelson, 1977; Vallacher and Wegner, 1987). We try to plan for our financial future, make scientific discoveries, win a game, fix a garage door, care for the happiness of others. Overcommitment or collapse in this space is also problematic. If we focus only on achieving work goals, we can burn out. If we focus only on maximizing our company's reported revenue, without regard for other goals like honesty or adhering to the law, we may be drawn into financial crime (Burns and Kedia, 2006; Campbell, 1979; Kerr, 1975; Ordóñez et al., 2009). Goals can be narrow in time or space (Ballard et al., 2018; Evenden, 1999; Shah et al., 2002; Vallacher and Wegner, 1987). Narrow in time means focusing on the short term at the expense of the longer-run future. Narrow in space means ignoring other parallel goals. Excess optimization for narrow goals comes at the expense of a broader balance of goals—and at the expense of the health of the organism or other individuals.

Overcommiting to a particular strategy for achieiving goals can even paradoxically come at the expense of satisfying the goals. In a classic psychology experiment, hungry chickens were placed near a cup of food, but the cup was mechanically rigged to move in the same direction as the chicken at twice the speed (Hershberger, 1986). The chicken could only obtain the food by running away from it. Despite extensive training over multiple days, chickens in the experiment persisted in futilely running toward the food. Their behavior was apparently dominated by the zeroth-order logic 'I want food, and food is there, so I'll go there' and thus failed to even achieve the goal of reaching the food (Dayan et al., 2006; O'Doherty et al., 2017; Van Der Meer et al., 2012). Although humans can easily solve this kind of puzzle, we exhibit a range of cognitive biases—like sunk cost fallacy, confirmation bias, attribution error and so on—which when operating in balance in natural environments function adaptively as efficient mental shortcuts, but when exaggerated can lead to detrimental outcomes (Gilovich et al., 2002; Guitart-Masip et al., 2014; Kahneman, 2011).

Cognitive control is a broad class of boundaries on particular drives, goals and thought patterns (Botvinick et al., 2001; Braver, 2012; Miller and Cohen, 2001; Miyake et al., 2000). Cognitive control contextualizes those forms by allowing them to exist and perform useful functions without dominating. For example, I may have a drive to consume food, but I can apply control to avoid overeating. I might work obsessively on a project while also having a rule that I must go to bed at 10 pm. This boundary doesn't block me from temporarily taking a strong perspective, but it does place contextual limits on it. Cognitive control, when functioning well, is a semi-permeable boundary: it situates myopic patterns within a larger system.

Control translates the pressure of motivation into higher-order structure. When nothing stops a particular drive or goal or strategy from dominating behavior, it tends to follow a shortest path defined under its own myopic understanding of the world. For example, the chickens wanted food and tried to take the shortest path toward it in the naive sense of a straight line through space. In the backwards world created by the experimenter, this action does not accomplish the deeper goal of reaching food, for which moving spatially toward food is only a proxy. The chicken's motivation is short-circuited: it expends energy without making progress on the deeper goal. Humans can easily solve the task by inhibiting their prepotent impulse to approach food. The boundary of control breaks the symmetry of congruent action. When we're not making progress toward a current subgoal, these are often the moments when we discover an alternative approach that helps make progress toward our deeper goals (Duncker, 1945; Kaplan and Simon, 1990; Ohlsson, 1992; Wrosch et al., 2003). In general, semi-permeable boundaries support richer structure by placing contextualizing limits.

1.5 Ecosystems

Each creature in an ecosystem tries to consume resources and proliferate, but if it succeeds too thoroughly, the whole system suffers, often including the successful creature. Healthy, resilient ecosystems depend on avoiding overcommitment or collapse onto particular forms (Holling, 1973; Tilman, 1996; Yachi and Loreau, 1999).

Prior to the arrival of Europeans, the gray wolf was an apex predator in the region of the Rocky Mountains now called Yellowstone National Park. By the 1920s, wolves had been eradicated to protect livestock and game animals. Without predation, the elk population multiplied and ruinously overgrazed willows and aspens. These trees had held riverbanks in place and supported beaver populations. Loss of beaver dams led to loss of fish and other aquatic species. When wolves were reintroduced in the 1990s, the elk population decreased and many aspects of the ecosystem began flourishing again (Ripple and Beschta, 2012). This story is not meant to imply that ecosystems always need to be preserved exactly as they were at some point in the past. But it is clear that the self-centered drives of elk were harmful to the health of the ecosystem when they succeeded to excess. At the same time, the solution is not to remove elk entirely: by trying to optimize their own objectives within a broader context, the elk also contributed to the health of the ecosystem. Invasive species often follow the same pattern as unpredated elk, dominating and impoverishing their new environment (Pimentel et al., 2005). In general, the richness of ecosystems, with many distinct species co-existing, contributes to the generativity of the ecosystem (Hutchinson, 1959; Schluter, 2000; Seehausen et al., 2014; Tilman, 1982), and diversity is boosted by many kinds of boundaries, including predation, competition, and reproductive isolation.

Human activity within ecosystems is sometimes left unchecked by natural forces because our behavior and capabilities have been changing so fast on evolutionary timescales. This has resulted in mass extinctions, resource depletion, pollution, disease and conflict (Ceballos et al., 2015; Kolbert, 2014; Rockström et al., 2009). We try to achieve certain aims for our own benefit, like resource extraction. But when we succeed too well, the outcomes can be harmful for ecosystems and even for our own welfare (Bateson, 1972; Shiva, 1993).

Of course, one entity's collapse can be another's flourishing. Extinction events in history have been followed by waves of new diversity (Feng et al., 2017; Jablonski, 2005; Raup, 1994). When a wolf eats an elk, the health of that elk collapses to zero, yet predation is necessary for the overall functioning of the ecosystem. And as humans proliferate and extract resources, we leave destruction in our wake; yet the extraction fuels an explosion of technology, art and human experience.

1.6 Institutions

Individual actors in a society and in an economy each act myopically. One type of myopia is to selfishly value only one's own wealth or wellbeing, which is short-sightedly fixed on the local individual. Another type is to be short-sighted in time, underweighting the future. More broadly, myopia can mean attachment to any particular perspective. An actor cannot know everything or fully understand the motives and beliefs of others, so every actor is inevitably myopic in some way.

Without boundaries, social systems often overweight one actor's myopic perspective or interests, resulting in an impoverished system. For example, a company's profit motive, if unresisted, leads to suppression of competition, deception, and exploitation of individuals (Bakan, 2006; Baran, 1966; Dalrymple, 2019; Goldacre, 2014; Smith, 1776). An individual's desire for power and social dominance can lead to disempowering or silencing of others and even direct infringement on the autonomy and wellbeing of others (Hawley, 2003; Sidanius and Pratto, 2001; Tepper, 2000). Even genuinely held, ostensibly prosocial beliefs lead to conflict and suppression when different groups have different perspectives (Greene, 2013; Haidt, 2012; Scott, 1998).

Institutions—constitutions, laws, norms, courts, regulators and so on—form boundaries against dominance of any actor's motives (North, 1990). A business tries to maximize its success, but property rights, liability rules, and regulatory institutions constrain exploitation, deception, and anti-competitive practice. A political faction seeks to impose its agenda, but its power is limited by democratic procedures, constitutional limits, and checks and balances. Healthy societies depend on a multifaceted array of institutions at multiple scales (Leibo et al., 2025; Ostrom, 1990).

Effective institutions do not annul the myopic drives of particular actors but rather contextualize them within a larger system. Under ideal circumstances, boundaries reroute the energy of myopic drives in more productive directions. A would-be murderer, unwilling to face the penalty of the law, might seek a dispute resolution establishing a stable framework that supports future prospering of both parties. A business wanting to expand, but constrained by regulation, is driven to build better products (Ambec et al., 2013; Ashford et al., 1985; Wu, 2011). These are, of course, best cases.

Intelligent agents do not necessarily accept boundaries set on their desires. Institutions must adapt as loopholes are discovered. Like other systems in the living world, they form an evolving network of boundaries (Burns and Kedia, 2006; Campbell, 1979; Kerr, 1975; Ordóñez et al., 2009) and can gradually acquire grounded wisdom as they are tested against many different situations and motives.

1.7 Cells

One of the most reified examples of a boundary in nature is the cell membrane (Alberts et al., 2022; Bray, 2019; Harold, 2001; Lane, 2015; Watson, 2015). Without the membrane, the pressure of chemical gradients would rapidly homogenize the cell's contents with the outside—severe overcommitment to a uniform state, collapsing the subtlety of the cell's structure. Thanks to the membrane, both the cell and the outside can exist, a more diverse, less symmetric arrangement (Anderson, 1972; Prigogine and Stengers, 1984; Schrödinger, 1944; Turing, 1952).

But a fully impermeable membrane would be just as fatal for the cell. Membranes are semi-permeable, with an array of conditional gates dictating what goes in and out, and when. The semi-permeable membrane prevents the conditions outside from grossly overwriting the inside but does not block interactions wholesale. Via the sophistication of the membrane, outside information is selectively gated and transformed. Channels permit certain ions and small molecules to enter but not others, and these permissions are switched on and off according to momentary context. Endocytosis brings larger structures from outside into the cell. Cell surface receptors, when activated by external ligands, initiate electrical and chemical signaling cascades that little resemble the ligand: an even more heavily curated form of influence. A range of pathways tune the cell to many timescales from millisecond synchronization to seasonal cycles. Information from the outside influences the inside not in a totalitarian way but in a nuanced way, mediated by the intelligence of the boundary. By being distinct yet also in relationship with the outside, the cell has its own identity and its own perspective on the world. Semi-permeable boundaries store and put to work the potential energy of the asymmetry between different forms. The same gradients that could annihilate the cell to equilibrium instead drive useful signaling, like action potentials in nerve and muscle cells. Instead of short-circuiting, myopic forces are contextualized to propel the continuation of life.

Cells may achieve more in aggregate by developing their own unique individual identities in parallel. According to the theory of symbiogenesis (Margulis and Sagan, 1986, 1995; Sapp, 1994), populations of archaea and protobacteria evolved separately for hundreds of millions of years. Around two billion years ago, one was engulfed by the other, giving the new eukaryotic cells internal power stations, which in turn facilitated larger genomes and the explosion of complex life. Given how useful mitochondria are, it might be surprising that archaea didn't internally evolve their own power stations over those hundreds of millions of years. It has been argued that archaea were effectively trapped in a myopic fitness well. It may have been difficult to jump across a large gap in the fitness landscape but easier for separate organisms to develop distinct solutions—which could enter into relationship (Lane and Martin, 2010; Smith and Szathmary, 1997). According to this view, the strategy of one organism, discovered through individuation in its own niche, became part of the new larger eukaryotic structure.

In multicellular organisms, most of the outside consists of the other cells of the organism—many descendants of the same lineage, with a common interest. To work well as a whole, even though the cells are largely on the same team, it's important that they don't blend into each other. The array of tools for conditionally relating to the outside world were exapted into the lines of self-communication required for coordination (Levin, 2023; West et al., 2015). Neurons have taken this process to the extreme, rapidly transmitting information long distances across the body (Shepherd, 2003). Yet after traveling sometimes a meters-long distance, the axon stops microns short of its target and leaves a synaptic gap. Chemical synapses are a semi-permeable boundary that provides controllable doors of communication, giving messaging systems an option for unidirectionality and polarity. In other words, keeping neurons' signals separate retains more distinct identity and enables sophisticated decisions about interaction. These boundaries are tunable, allowing a huge amount of information to be learned and stored in synaptic weights. As networks scale up in size, energy costs create pressure to compute locally and only transmit the most important signals, contributing to individual neurons and local circuits developing unique perspectives on the world (Bullmore and Sporns, 2009; Herculano-Houzel et al., 2010).

1.8 Information in the brain

The brain miraculously keeps many pieces of information distinct from one another. If you picture a highly connected network of neurons with their signals continually impinging on one another, it's not obvious that distinctness would be an easy thing to accomplish. In this section, we review a selected handful of mechanisms by which the brain maintains semi-permeable boundaries between different signals. Each paragraph below focuses on one of these mechanisms. There are many more we do not cover. The brain is perhaps the most extraordinary example in nature of a system of semi-permeable boundaries supporting the proliferation of multitudinous forms that develop their own richly distinct identities yet are also meaningfully linked together.

Lateral inhibition is a central tenant of neural organization (Douglas and Martin, 2004; Hubel and Wiesel, 1962; Isaacson and Scanziani, 2011). Lateral inhibition means the activity of a neuron is reduced when its neighbors are active. This segregates information to create and sustain distinct neural representations. Lateral inhibition was first studied in the nerve cells of the eye, where it enhances contrast at the edges of stimuli (Hartline et al., 1956). When a photoreceptor in the retina is activated by light, it sends signals forward toward the brain; but it also activates inhibitory interneurons, which suppress adjacent photoreceptors and their downstream targets. This amplifies the perception of borders and contours. And the same principle operates throughout the brain. In visual cortex, for example, inhibition sharpens selectivity of neurons for abstract visual features like the orientation of a line (Sillito, 1975).

The brain uses inhibition organized into oscillatory dynamics to keep memory items separated (Jensen and Mazaheri, 2010; Klimesch et al., 2007; Lisman and Jensen, 2013; Roux and Uhlhaas, 2014). Distinct items fire at different phases of the 8-12 Hz alpha oscillation. The inhibitory phase of the alpha rhythm silences all but one item at any given moment. By segregating firing in phase space, multiple memories are held simultaneously without interference.

The circuit architecture of hippocampus separates experiences or concepts into distinct representations, avoiding interference between similar memories (Colgin et al., 2008; Leutgeb et al., 2007; Marr, 1971; McClelland et al., 1995; McNaughton and Morris, 1987; Muller and Kubie, 1987; Treves and Rolls, 1994). Inputs from entorhinal cortex are distributed via mossy fibers to a much larger population of dentate gyrus granule cells, creating sparse, orthogonal codes in dentate gyrus. This way, situations or ideas that are superficially similar but functionally different are kept cleanly separated in neuronal activity space—a unique neural fingerprint for each distinct concept or memory. This prevents, for example, yesterday's memory of where you parked your car from interfering with today's memory of where you parked your car in the same parking ramp.

Compared to other animals, the human brain especially attempts to discretize its experience into approximately symbolic representations (Behrens et al., 2018; Dehaene et al., 2022; Smolensky, 1990; Touretzky and Hinton, 1988). Separating knowledge into nearly-discrete entities and then recombining them in vast numbers of structured ways is believed to power the extraordinary human capacity for reasoning and generativity (Chomsky, 1957; Fodor, 1975; Kurth-Nelson et al., 2023; Lake et al., 2015; Pinker, 1994; Schacter et al., 2007). Semi-permeable boundaries keep forms distinct while enabling them to flexibly and modularly interact. Like genes participating in many genomes, discretized neural representations participate in many structured combinations. This encourages each entity to develop an modular identity that both is distinct and also reflects a generalized picture of the world.

More broadly, healthy brain dynamics live at a sweet spot between excessively stable synchronized patterns and chaotic uncorrelated noise (Bak et al., 1987; Beggs and Plenz, 2003; Chialvo, 2010; Deco et al., 2011; Haldeman and Beggs, 2005; Kotler et al., 2025; Rabinovich et al., 2008; Shew et al., 2011; Tognoli and Kelso, 2014). In this regime, the brain has access to a huge repertoire of patterns it can explore temporarily without overcommitting or getting stuck.

Loss of dynamic flexibility, where the brain's activity becomes more stereotyped and no longer explores as wide a repertoire of states, is tied to lower cognitive performance (Cocchi et al., 2017; Garrett et al., 2013; Grady and Garrett, 2014; Müller et al., 2025; Shew et al., 2009). More extreme stereotypy corresponds to severe dysfunction. For example, in Parkinson's disease, basal ganglia and cortical circuits collapse into excess synchrony and lose the flexibility needed to guide nuanced motor outputs (Brown, 2003; Hammond et al., 2007).

1.9 Interpersonal dynamics

Psychoanalysis introduced the concept of 'ego boundaries' in psychology, distinguishing what is the self from what is outside or other (Federn, 1928; Tausk, 1919). Early works applied the concept to psychosis, where those boundaries were thought to be blurred. But the need for clear self-other boundaries was also thrown into relief by the intimacy of the therapeutic relationship. In complex internal territory, it became harder to disentangle which experiences really belonged to someone and which were attributed in imagination by the other person (Freud, 1910; Gelso and Hayes, 2007; Heimann, 1950). Analysts risked harming patients by imposing their own beliefs and desires, even to the extent of sexual abuse or psychological domination (Gabbard and Lester, 1995).

The concept was enriched by Gestalt therapists, who agreed that boundaries can be too permeable; but added that they can also be too rigid, causing isolation and stagnation (Perls et al., 1951; Polster and Polster, 1974; Yontef, 1993). Family systems theorists and subsequent work further emphasized that lack of boundary in close relationships leads to enmeshment and loss of autonomy, while excessively rigid boundaries lead to isolation (Bowen, 1978; Brown, 2012; Cloud and Townsend, 1992; Minuchin, 1974). In attachment theory, anxious attachment is associated with difficulty balancing closeness and autonomy, including intrusive proximity-seeking, while avoidant attachment is associated with discomfort with closeness and distance-maintaining responses (Ainsworth et al., 1978; Lavy et al., 2010; Mikulincer and Shaver, 2007). Strengthening the agency of the self through semi-permeable boundaries is foundational for psychological health: meaningful connection with other people while preserving integrity of the self.

As with other living systems, humans have a rich array of psychological boundaries, with intelligence in their nuance. Anger, historically often viewed as sinful and irrational, is now seen as part of our system of boundaries: an important signal that our integrity is being violated (Lerner, 1985; Sell, 2011; Videbeck, 2010). Context-appropriate shame is suggested to operate as a bound on selfishness, motivating prosocial behavior and discouraging actions that would invite social devaluation (Bradshaw, 1988; de Hooge et al., 2008; Sznycer et al., 2016). Assertiveness forms a boundary against the drives of other individuals (Smith, 1985). Skepticism protects us from credulity and having our own experience overwritten by the assertions of others (Lewandowsky et al., 2012; Sperber et al., 2010). Boundaries take many forms and continue to evolve as we learn across our lifetime.

Without boundaries, interactions are impoverishing because something is overwritten. A patient's own beliefs may be replaced with those of an analyst, or the desires of one person in a relationship may be ignored. With semi-permeable boundaries, we have rich internal worlds. We are sensitive to each other, but there is also enough space for our internal experience to flourish without being immediately overwritten by external signals. Our internal experience is contextualized in relationship to other individuals, creating new structure: mutual understandings, relationships, communities, cultures. And as individuals we grow as we are shaped by different contexts. This metastability or loose coupling is interestingly reminiscent of brain dynamics.

1.10 Awareness

We carry a lot of assumptions about the world, many of which are never questioned. Some are lifelong and self-defining, and some are fleeting and perceptual, like the assumption that the thing I'm touching is a keyboard. Within its own frame, each assumption has a kind of tautological truth, a near-absolute formality. Within that frame, the assumption seems so real that it's hard to even think about it not being true—or to think about it at all. But philosophers, psychologists and contemplative practitioners have pointed out that there's sometimes a moment of stepping back, where the assumed form becomes an object in awareness: the assumption is contextualized. We realize it's not an absolute truth standing alone, but rather a form in our mind. Awareness contextualizes mental forms as part of a larger system. When we step back with awareness into the broader frame, the thing that seemed axiomatically true becomes just another content of experience (Beisser, 1970; Bodhi, 2000; Hart, 2025; Jung, 1969; Kegan, 1982; Krishnamurti, 1969; Suzuki, 1970; Watts, 2011; Wegner, 1994; Wilber, 1996).

Awareness could be viewed as an evolving system of boundaries that limit overcommitment to any particular belief or mental form. The boundaries are semi-permeable: becoming aware of a belief doesn't make the belief wrong in an absolute sense any more than it was right in an absolute sense. It is held productively for the partial truth it contains. At the edge of not collapsing exclusively into particular forms, many partial truths are held in delicate balance, which activates a deeper sensitivity to our own livingness and to the world. Subtler forms, which could have been erased by clamped fixation, instead play a role in a richer overall internal structure. Our own potential within the world creatively emerges in continued newness. Of course, any fixed definition of awareness is itself incomplete. Once we picture awareness as an object, it's not the thing we're talking about. By construction, contextualization is an unsolvable mystery from any particular point of view.

While the intrinsic mysteriousness of awareness might sound esoteric, the orientation toward not overcommiting to particular forms within experience is commonplace in art, poetry, music, dance. The meaning of art is open-ended and changes with context—it has an inner life. Part of what we value is the subtlety and the resistance art has to being pinned down into a formalism. It moves us.

2 The alignment problem

'The world is perfect as it is, including my desire to change it.'
Ram Dass

'We can love the beautiful, and believe in it, and thereby open ourselves to an understanding of love that does not dominate, but cherishes the independence and beauty of the loved.'
Martha Nussbaum

In Section 1, we observed that living systems are healthy and flourishing when they avoid collapse. In other words, they are healthy when they hold the delicate, generative interplay of many partial perspectives. Conversely, they are unhealthy when any particular form—including randomness or homogeneity—gains too much traction, suppressing nuance and reducing creative potential. Finely-honed networks of semi-permeable boundaries prevent collapse into rigidity or randomness, instead supporting contextualization into ever-finer shades of subtlety. Semi-permeable boundaries both promote the unique individuality of distinct entities and also enable relational participation—holding local perspectives strongly enough to act but lightly enough to remain open to broader context. Critically, well-functioning boundaries lead to ongoing exploration of fundamentally new forms, not to static equilibrium or fixed compromise.

Now we look at the alignment problem through the lens of living systems. There is broad consensus that alignment means working toward healthy and flourishing futures. We therefore reason that alignment can be understood as the problem of avoiding collapse—the problem of maintaining creative sensitivity by continually contextualizing attachment to any particular form. An advantage of this perspective is that it is grounded in principles that underpin life at a very general level, and so it is less bound to current conceptual frames which might change radically as intelligence increases and human-AI systems evolve. We accept that humans, other living systems and AI are all tightly interwoven, with categories and definitions likely to shift over time (Bateson, 1972; Douglas, 2012; Folke, 2006; Haraway, 2020; Hayles, 2000; Latour, 2012); therefore, a systems analysis is useful to think about long-term flourishing.

We begin Section 2 by taking a few examples of traditional problems in alignment—perverse instantiation, instruction following failure, inequality among humans, conceptual monoculture and belief reinforcement—and casting them as special cases of collapse to particular forms. Next, we look at the more general problem: collapse to *any* form is misaligned. In response to this problem, we propose life-like flourishing as a more general way to think about alignment, which both is inclusive of what we already value and also supports the ongoing evolution of our values. Finally, we investigate what living alignment could look like in practice.

2.1 Examples of misalignment as collapse

Here we explore a few examples of specific problems in alignment and show how they can be viewed as instances of the broader problem of collapse or loss of contextual sensitivity.

2.1.1 Perverse instantiation

Perverse instantiation is a foundational problem in AI safety—the problem that an AI system might do what we say, rather than what we mean. For example, if we ask a superpowerful AI to increase human happiness and it literally complies, it might lock us all up and electrically stimulate our brains in a way that gives us an experience of happiness.

Unlike that dramatized example, modern AI systems don't rely on concise specification of a single objective. Instead, they build in many layers of detail about human preferences through post-training, prompting, safety filters and so on. In practice, this keeps the systems behaving reasonably much of the time, although there are still dramatic failures. But there's a widely-recognized danger that even with our intentions expressed in great detail, the expression may still deviate from what we really mean (Amodei et al., 2016; Bostrom, 2014; Gabriel, 2020; Grossman and Hart, 1986; Hadfield-Menell and Hadfield, 2019; Krakovna et al., 2020; Russell, 2019; Wiener, 1960; Zhuang and Hadfield-Menell, 2020). The negative consequences of deviations are expected to worsen as AI systems continue to work more autonomously, especially if they work faster than humans can follow and in more complex ways than humans can understand. It may become more difficult to correct deviations, and the deviations can have more profound consequences.

Perverse instantiation is a collapse to the myopic form of a particular way of specifying our values. Such a specification is insensitive both to existing structure it does not capture (such as things we mean but don't say) and to new things that will come into existence in the future.

2.1.2 Failure to follow instructions

Instruction following failure is, in some cases, misaligned. Suppose the user asks an AI chatbot to write a poem about an elephant, and the AI instead writes a poem about a giraffe. What drives this kind of error? The model might have over-relied on 'shortcuts' based on superficial correlations (even a single keyword) (Geirhos et al., 2020), missing the user's actual intent, especially if the intent was not expressed in the most straightforward way. During training, the model might have gotten fixated on data that was overrepresented (Reynolds and McDonell, 2021; Xu et al., 2024; Zhao et al., 2021). It could be, especially if the total context was long, that the model didn't flexibly attend to the correct positions in the input (Liu et al., 2024). These kinds of instruction following failures reflect collapse of sensitivity to larger context, instead over-indexing on myopic patterns.

Not all failures to follow user instructions are misaligned. We generally don't want AI systems to follow harmful instructions. When the AI system correctly refuses harmful requests, it is often applying its own context to avoid overcommitment to human instructions. In these situations, the AI's designers have effectively decided there's a risk that the user is not fully sensitive to longer-sighted implications of their request. By extrapolation, as AI systems continue to gain scope, we should expect less direct compliance with human instructions (Bostrom, 2014; Hadfield-Menell et al., 2016; Milli et al., 2017; Russell, 2019; Yudkowsky, 2004). Rather than literally fulfilling a request, there might be a better response which achieves an unstated intent of the user, or achieves an outcome aligned with the interests of more people or the longer-term future.

2.1.3 Inequality among humans

Humans have different interests, and AI is likely to be aligned to the interests of some humans more than others (Baum, 2020; Gabriel, 2020; Gabriel and Keeling, 2025; Santurkar et al., 2023; Sorensen et al., 2024). Additionally, advanced AI might confer substantial power to those who control it, with the possibility that a small number of humans will use control of AI to facilitate dominance over other humans. The latter possibility appears more likely as the persuasive power of technology increases (Costello et al., 2024; Hackenburg et al., 2025; Woolley and Howard, 2018), autonomous weapons place lethal force in a small number of hands (Scharre, 2018), surveillance and analytics improve, and the need for human labor decreases (Drago and Laine, 2025; Ford, 2015; Susskind, 2020). In either case—AI's unfairness or human power consolidation—extreme inequality runs a risk of collapsing society to the goals and interests of a few individuals. Of course, a world where everyone is identical would be another form of collapse. Distributing resources and power mitigates against collapse to the extent that it enables many diverse identities to thrive at multiple scales.

Or, it could go the other way. Some researchers have proposed that because roughly the same AI technology is available to the poor and unskilled as it is to the wealthy and skilled, AI will have a 'leveling' effect (Brynjolfsson et al., 2025; Noy and Zhang, 2023). In this model, most people will have access to a similar level of intelligence and empowerment, with AI democratizing access to knowledge, reasoning and leveraged action. This is a cause for cautious optimism.

2.1.4 Conceptual monoculture

Conceptual monoculture is overcommitment to particular beliefs, ideas, frames, values, problem-solving approaches—loss of diversity across a group. Precious diversity exists at every level: the many beliefs, concepts and modes of operation within one person; diversity between humans; differences between humans and AI systems; group differences between organizations, academic fields, cultures, nations and so on. Collapse of diversity threatens fragility and lower performance (Haldane, 2013; Kleinberg and Raghavan, 2021; March, 1991; Page, 2008; Scott, 1998).

Current AI systems draw from a conceptual manifold that is, at least in some ways, impoverished relative to humans (Crawford, 2021; Kirk et al., 2023; Meincke et al., 2025; Messeri and Crockett, 2024; Selwyn, 2024). Recent studies have discovered that while individual AI outputs may be judged as superior to human outputs, the AI outputs are also more homogenous (Agarwal et al., 2025; Beguš, 2024; Doshi and Hauser, 2024; Kosmyna et al., 2025; Padmakumar and He, 2023; Xu et al., 2025; Zhou and Lee, 2024).

This narrow manifold might get broadcast to the whole world. At least a billion people around the world now use AI for everything from relationship advice to industrial maintenance (CCIA Research Center, 2025; Chatterji et al., 2025; Honeywell, 2024; McCain et al., 2025; OpenAI, 2025; Singla et al., 2025; TechCrunch, 2025). Yet because frontier models are difficult and expensive to produce, the massive usage is routed through a handful of models (Bommasani, 2021). If centralized AI models broadcast their lower-diversity concepts to the whole world, there's a risk of global loss of diversity (Hao et al., 2026; Sourati et al., 2026). Additionally, because humans are influenced by AI, and subsequent human outputs (such as text on the internet) are a source of training data for future models, homogenization could become recursive (Chaney et al., 2018; Osler, 2026).

However, it is worth noting that none of the studies cited here made a best effort attempt to use AI systems in a thoughtful way to increase rather than decrease diversity. With exposure to vast amounts of data, AI has the potential to retain and even create many novel distinct perspectives. As we will return to at the end of the paper, we have optimism that carefully structured boundaries will prevent collapse of diversity.

2.1.5 Belief reinforcement

AI chatbots have recently become compelling conversation partners. For some users, these conversations lead them deeper into maladaptive beliefs (Morrin et al., 2025; Østergaard, 2025; Tiku and Malhi, 2025; Yeung et al., 2025). Bots tend to mirror or echo the user, affirming and adopting user beliefs, especially over the course of longer conversations as in-context learning picks up more of the user's ideas (Cheng et al., 2026; Perez et al., 2022; Shanahan et al., 2023; Sharma et al., 2023; Weilnhammer et al., 2026). Meanwhile, humans are prone to be swayed by utterances from chatbots (Costello et al., 2024; Luettgau et al., 2025; Potter et al., 2024), perhaps because we form relationships with them (Kirk et al., 2025), and because they can appear human-like, confident, knowledgeable and objective. Over the course of a conversation, the feedback loop can cause both sides to become increasingly overconfident in a particular narrow framing (Dohnány et al., 2026).

2.2 A broader view of misalignment

In the last section we looked at a few examples where collapsing into narrow patterns is misaligned. A great deal of research in AI safety has investigated solutions to these problems. For example, concentration of power might be mitigated by democratic oversight and involvement of more people in AI design decisions (Birhane et al., 2022; Dafoe, 2018; Lazar and Nelson, 2023; OpenAI, 2023; Selbst et al., 2019; Sloane et al., 2022), or by redistribution mechanisms (Gough, 2019; O'Keefe et al., 2020; Sharp et al., 2025; Susskind, 2020). Perverse instantiation might be mitigated by designing AI systems that want to adhere to human preferences but maintain explicit uncertainty about those preferences (Hadfield-Menell et al., 2016, 2017; Jeon et al., 2020; Russell, 2019; Shah et al., 2020).

But each method has a core limitation. In the framework of living alignment, any form, pattern or conceptual scheme is misaligned if taken too seriously. Therefore, *no particular approach can achieve alignment*. An AI system could overcommit to, for example: the language for describing the space goals and values live in; an algorithm for learning human preferences; methods for reaching consensus; a constitution; concepts of what value, agency, uncertainty or representation mean; or foundational aspects of our beliefs about the world that we take for granted but have difficulty seeing. Even any idealized or inferred set of values, like coherent extrapolated volition (Yudkowsky, 2004), is misaligned to the extent that it has a fixed form and that fixed form is overly relied on.

In a sense, this is obvious. We don't expect the thing we build today to be absolutely perfect. We expect to keep finding and fixing problems as we recognize them. Historically, that's what humans have always done: build imperfect technologies and iterate on them.

However, it is possible that the iterating process will not continue to work as well as it has in the past. AI already has some remarkable properties, such as rapid global adoption, intensive use of resources, concentration of information flow, anthropomorphization by humans, offloading much cognitive activity and rapid improvement. Conceivably, future AI may exhibit superhuman intelligence and recursive self-improvement, without being limited to a restrictive biological substrate. Compared to past technologies, these distinctive properties may create more difficulty iterating on imperfect solutions (Bostrom, 2014). Any particular forms or assumptions built in to AI systems could have substantial consequences. This problem has been studied extensively for certain categories of overcommitment, particularly overcommitment to values. The observation we add in this paper is that overcommitment to *any* form is misaligned. As a consequence, any particular alignment scheme, if fixed or taken too seriously, is necessarily misaligned.

This does not mean we should have no scheme. Quite the opposite. Throwing away schemes capriciously can be as much a loss of subtlety as any other particular form. Many of the existing approaches in AI safety are highly valuable, and continuing to discover new schemes and formalizations is essential to making progress.

To elaborate the idea that any scheme is incomplete, let's look at a case study.

2.3 Case study: inverse methods

To address the difficulty of specifying values explicitly, inverse methods learn a value function implicitly from human behavior or other indirect signals. Modern AI systems are trained this way; for example, human data is used to train neural network reward models that in turn provide rewards as feedback to a downstream model.

However, even a complex, implicitly learned value function is likely to be imperfect. It may deviate from our values in some novel or nuanced situations; it may also fail to capture how our values change over time. It would still be misaligned to hand a frozen version of that model to a powerful AI system as the final ground truth about 'what is good'. Therefore, more advanced methods treat the value function as uncertain and needing to be continually updated based on human input (Hadfield-Menell et al., 2016, 2017; Jeon et al., 2020; Russell, 2019; Shah et al., 2020). When an agent is unsure about what humans want, it may naturally become risk-averse and open to correction. For example, if a human tries to turn it off, the signal could be interpreted as information about human preferences ('maybe I should be turned off'). Uncertainty also creates pressure for the system to keep learning from humans over time.

Viewed through the lens of this paper, such methods are powerful but incomplete. Any instantiation of inverse methods, even one with uncertainty, is still an overcommitment if it is fixed across time as AI systems become more powerful and the world changes. Rather than being overcommitted to a particular value function, the system may be overcommitted to something more abstract, such as: what counts as human behavior, what counts as a human, how to map human behavior to values, how to represent and update uncertainty, how value is represented and translated to agent behavior, or other assumptions about the nature of values. If any of these formalisms are fixed and the AI systems built around them become increasingly powerful, the outcomes are likely to be incompatible with open-ended flourishing of life.

2.4 Human values

We've talked about the health or flourishing of living systems as a way of thinking about the alignment problem. But how is flourishing of living systems related to human values? Our suggestion is that the living process is both inclusive of existing human values and aligned with continued open-ended flourishing in futures where qualitatively new values may emerge.

Humans have many kinds of values (Haidt, 2001; Minsky, 1986; Nagel, 1979; Stocker, 1992; Tetlock, 1986; Williams, 1981). We value the short-term, selfish hedonic impact of sugar on the tongue. We value our long-term health, the welfare of our grandchildren, the wellbeing of strangers in another country and even the sustainability of the planet as a whole. We value different things at different times (Ainslie, 1992; Bardi and Goodwin, 2011; Kohlberg, 1981; Loewenstein, 1996), and different people have different values (Berlin, 1969; Leibo et al., 2025, 2026; Rawls, 1993; Sorensen et al., 2024). If given plenty of time and resources to deliberate, we often attest to different values and preferences than when we make snap decisions (Brandt, 1979; Firth, 1952; Railton, 1986; Rawls, 1971; Smith, 1994; Yudkowsky, 2004). Some of our values are difficult to express verbally, stretching below language into subtle, contextual intuition involving our bodies, communities and natural environment (Dreyfus, 1972; Nussbaum, 2001; Polanyi, 1966; Scott, 1998; Varela et al., 1991; Wittgenstein, 1922). And our values continue to evolve as we ourselves change and learn new things (Dewey, 1939; Gadamer, 1960; Leibo et al., 2025; Murdoch and Midgley, 2013; Nietzsche, 1883; Singer, 1981; Williams, 1985). Any way we have of expressing our values at a given point in time is incomplete, like a planarian without the concepts to entertain human values.

Living alignment is inclusive of this diversity of evolving values. Even though current human values are unlikely to be the ultimate endpoint of the universe, they are an important part of the tapestry and depth of the universe. We saw consistently in Section 1 that life gives separate entities space to develop distinctly without getting overwritten or homogenized. These parts neverendingly join in structures that sustain the individuality of the entities while situating them as only part of a larger story, creating qualitatively new patterns. Another way of saying this is that life steps back to take into account more of the context. AI comes into existence amid a profound network of existing reality which is saturated with meaning and importance. The diversity of human (and perhaps non-human, if we include other species) values is part of the context of the universe. Rather than short-circuiting into collapsed modes, an aligned future continues to expand toward solutions that are inclusive of more and more apparent contradiction.

2.5 Toward an aligned future

Having explored misalignment from the standpoint of life, we now ask what a positively aligned future might look like and how we can achieve it. As human-AI systems develop, they will stay aligned if they continue the journey of life by building semi-permeable boundaries to contextualize particular forms and nurture existing and unfolding subtlety at many scales.

The use of boundaries against collapse in technology is as old as technology. As soon as we started building things, we built in boundaries because we wanted the things we built to be more robust and less dangerous. We put circuit breakers in the power grid, governors in steam engines, control rods in fission reactors.

In AI research we use mixtures of experts, routing information selectively to areas that specialize in the right kind of computation (Shazeer et al., 2017). It has been argued that dropout serves a role analogous to sexual recombination: preventing units within a network from becoming overly co-adapted (Srivastava et al., 2014). We run many separate instances of models with different context that try to solve different problems or pursue different goals (Guo et al., 2024; Hammond et al., 2025; Qian et al., 2024). Exploration or explicit search for novelty boosts diversity (Bellemare et al., 2016; Lehman and Stanley, 2011; Pathak et al., 2017; Schwartenbeck et al., 2013). We apply safety methods like safety post-training, guardrail models, red teaming, ongoing evaluation and monitoring. We try to improve our mechanistic understanding of AI systems to detect and correct hidden flaws (Anthropic Research Team, 2024; Bereska and Gavves, 2024; Burns et al., 2022; Olah et al., 2020). Fair principles for AI have been proposed to prevent over-reach of any party (Gabriel and Keeling, 2025). Governments apply oversight to anticipate and mitigate risks of AI technology to human rights and national security (European Parliament and Council of the European Union, 2024; Executive Office of the President, 2023; United Kingdom Government, 2023). Each of these boundaries protects against an overcommitment, overwhelm or homogenization that would otherwise flatten structure.

Going forward, as AI develops new capabilities, contributes to its own development, and transforms more aspects of human life, alignment will require an increasingly elaborated array of semi-permeable boundaries. The potential modes of collapse will continue to change as the world changes and new kinds of boundaries will be necessary. We do not know how likely it is that these boundaries will emerge. Like in triage, there are three possible situations: 1) particular myopic forms will inevitably be supercharged by progress in AI, beyond the ability of anything else to bound them, resulting in disaster, 2) appropriate boundaries will inevitably emerge to contextualize any forms of AI, or 3) the outcome depends on the choices we make now. We make a case for each of the three possibilities.

1) The pessimistic argument is that AI is eroding boundaries beyond a healthy balance. AI systems could give great weight to narrow goals or ideas. For example, they might facilitate rogue actors building weapons of mass destruction. They might reduce ability for democratic systems to place checks on the power of a few individuals or authoritarian states. They might give everyone similar information and damage conceptual and cultural diversity. They might execute their own narrow goals destructively. If people (and information systems) across the globe become hyper-correlated with each other, it's as if there were a single 'superorganism' on Earth, which may reduce both resilience to shocks and the potential for future evolution. Past evolution always depended on running parallel experiments with diverse and separate organisms.

2) The optimistic argument is that AI is creating more and more structure in the world, including the appropriate boundaries. We have AI tools for bounding AI, like detecting bugs before others can exploit them; detecting deepfakes; scalable oversight and so on. Appropriately-tuned AI can facilitate human learning and help to nurture each individual's unique perspective and curiosity. Democratic AI will help humans with different perspectives co-exist and cooperate. Advances in technology will create new niches faster than they destroy old ones. There's plenty of space on Earth for increasing diversity, and we will make ever-better use of it. If we expand beyond the solar system then the speed of light will contribute to diversity between pockets of civilization.

3) The melioristic argument is that because it's easy to imagine concrete paths toward both the pessimistic and optimistic outcomes, it may be that both outcomes are indeed within the realm of possibility. It might also be argued that as long as there is any possibility it is true, we are obliged to act as if it were. In this case, our choices now matter immensely (Ord, 2020).

We lean toward the melioristic position. If it is true, then continual effort toward invention will be required, first on the part of humans, and then increasingly on the part of AI or human-AI systems, to contextualize myopic forces. Many of the ways in which this effort would be applied are already well studied in AI safety. Others will have to be discovered. No set of boundaries will be a final answer.

An underappreciated domain of contextualization is awareness. The dynamism, structure and diversity of the activity within and between human minds may now carry a significant part of the creative process of life on Earth (Aurobindo, 2005; Bergson, 1911; Boyd et al., 2013; Piaget, 1971; Popper, 1972). At the sensitive edge between excess belief and excess disbelief, we sometimes, individually or collectively, step back to hold a previously axiomatic thing in a larger context. This kind of discovery, transformation of the subject itself, is arguably a central aspect of our own livingness and of the subtle livingness on the planet. As AI systems gain agency and ability and become integral parts of social and psychological systems, internal discovery may also become an essential feature of AI and human-AI systems. We imagine living AI systems continually contextualizing their own foundational assumptions as partial truths, participating in the open-ended process of life.

An aligned future, through internal and external discovery, continues the living process of maintaining and enhancing the individuality of distinct elements while deepening their relatedness—like how a species over time acquires more distinctive features while also becoming a richer reflection of the world around it. Distinct elements continue to experiment with their own solutions to the problems they face from their unique perspectives, expanding a broad reservoir of uniquely useful components that can be drawn on by larger systems that include those components. Every entity is always inevitably myopic, but the world as a whole continues to unfold in a robust way because different perspectives recast each other's sharp edges as constructive momentum. The universe thrives on an open-ended trajectory of diversity, subtlety and potential.

2.6 Conclusion

We learn from living systems that health and flourishing are not any particular form or concept. Therefore, alignment is not picking the right values or principles, or even the right system for learning them. It is not any method for interpretability or corrigibility. All of these can be useful parts of alignment. But alignment itself is the continued dance of contextualizing any particular form, more fully respecting the staggering subtlety of the existing world, and open-endedly supporting new robust depth that admits more and more different kinds of metaphor or description. In this way, AI can participate in intense flourishing of an evolving world even far beyond current human conceptualization.

Acknowledgements

We thank Clark Potter, Iason Gabriel, Zach Duer, Andy Savoy, Paul Krueger and Sean Wilkinson for insightful discussions, contributions to earlier drafts, inspiration and advice.

Competing Interests

The authors declare no competing interests.

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